

Lecture 5 – 23 September

Lecturer: Yu LI (李煜) from CSE

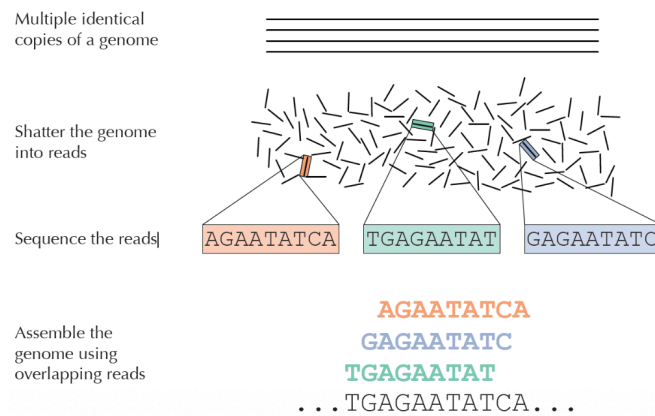
Scriber: Putu Juanita Arianti (1155199056)

Recap:

- Sequence data → data matrix
 - Map short reads to genome, then count number of reads → gene expression matrix
 - Genome seq from where? How to map?

Genome sequence assembly

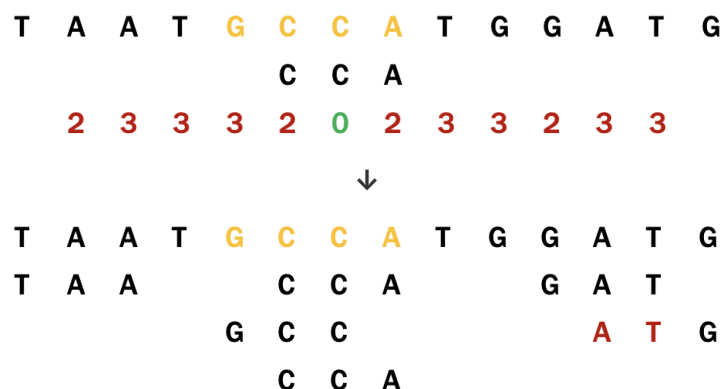
- Short reads can have overlap regions → genome assembly



Bioinformatics algorithms, 2015, Philip and Pavel

Mapping

- Slide each read along genome and calculate difference (no. of unmatching base pairs)
- e.g. for an **assembled genome** TAATGCCATGGATG against genome **reads** TAA, CCA, GAT, GCC, CCA, ATG
- **GCCA** is our target gene

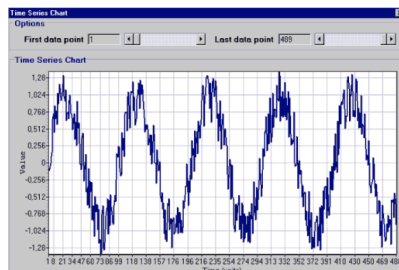


Gene expression count: 3

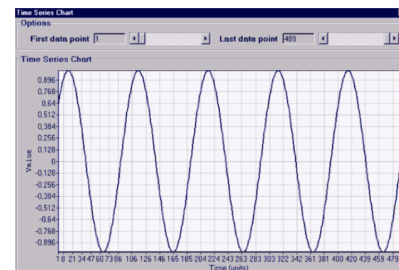
- We also see some reads e.g. ATG can be expressed in 3 different sites on the genome
- Any position is correct

Data cleaning

- We focus on cleaning data on **data matrix**
 - Collection of records with fixed sets of attributes
 - $n \times m$ matrix. n =object, m =attribute
- Data quality problems:
 - Noise and outliers
 - **Noise:** modification of original values, disruption/distortion affecting how clear we can identify original values
 - Cleaning by comparing point value against neighboring values
→ combine multiple values, avg, cancel out noise

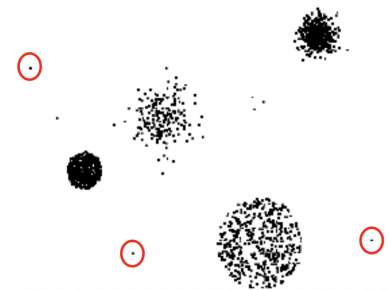


A sine wave with noise



The denoised sine wave

- **Outliers:** data points which are significantly different than most other data points in the data set
 - Clean by ignoring/not counting outliers



- Missing values

- Reasons:
 - Info **not collected** (e.g. people refuse to give age, weight)
 - Attributes **not applicable** (e.g. no annual income for children)
- Handling:
 - **Eliminate** data objects
 - **Estimate** missing values (with reference to most similar in dataset)
 - **Ignore** missing values during analysis
 - **Replace** with all possible values (weighted by probabilities)

Person	Height (m)	Weight (kg)
P1	1.79	75
P2	1.64	---
P3	1.70	63
P4	1.88	78

- Duplicate data (same set multiple times)
 - Big issue when **merging** heterogeneous sources (different databases)
e.g. same person using multiple emails
 - When merging the databases, if too many from one same person the data distribution will change → **inaccurate representation** of databases
 - Affect final result after statistical analysis

Database 1			Database 2		
Person	Height (m)	Weight (kg)	Person	Height (m)	Weight (kg)
P1	1.79	75	P1	1.79	75
P2	1.64	54	P7	1.65	55
P3	1.70	63	P8	1.69	63
P4	1.88	78	P9	1.87	77

- Unnormalized data
 - Attributes not on similar level of measurement
 - Height (m) vs weight (kg), from considering only the number itself weight is way larger
 - Diff distribution, diff value range → need to normalize within **same range of 0 to 1**
 - e.g. determine significance of expression of some genes in some samples
 - **Normalization:**

Person	Height (m)	Weight (kg)
P1	1.79	75
P2	1.64	54
P3	1.70	63
P4	1.88	78

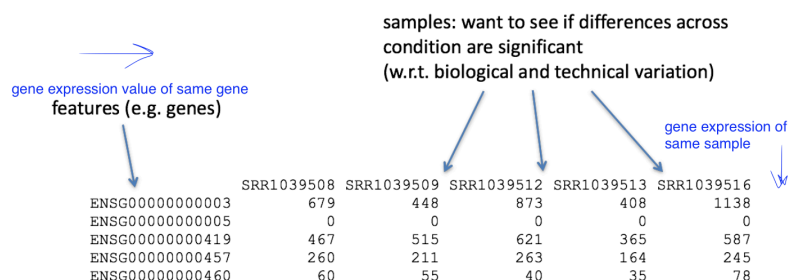
Min-max normalization

Person	Height (m)	Weight (kg)
P1	0.625	0.875
P2	0	0
P3	0.25	0.375
P4	1	1

- **Min-max:** $v' = \frac{v - v_{min}}{v_{max} - v_{min}}$
- **Z-score:** $v' = \frac{v - Mean(v)}{Std(v)}$, assuming Gaussian distribution

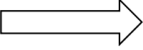
Normalize along **which direction?**

- With height & weight it's clear we want to investigate it in the same person, then compare between people
- With genes and samples we can either investigate gene expression of same gene within diff samples, or see diff gene expression within same sample → either works with proper rationalization



- Categorical data (descriptor, not numerical → difficult to find avg)
 - Computers handle numbers better
 - **One-hot encoding**

Person	Height(m)	Weight(kg)	Gender
P1	0.625	0.875	Male
P2	0	0	Female
P3	0.25	0.375	Female
P4	1	1	Male



Person	Height(m)	Weight(kg)	Male	Female
P1	0.625	0.875	1	0
P2	0	0	0	1
P3	0.25	0.375	0	1
P4	1	1	1	0

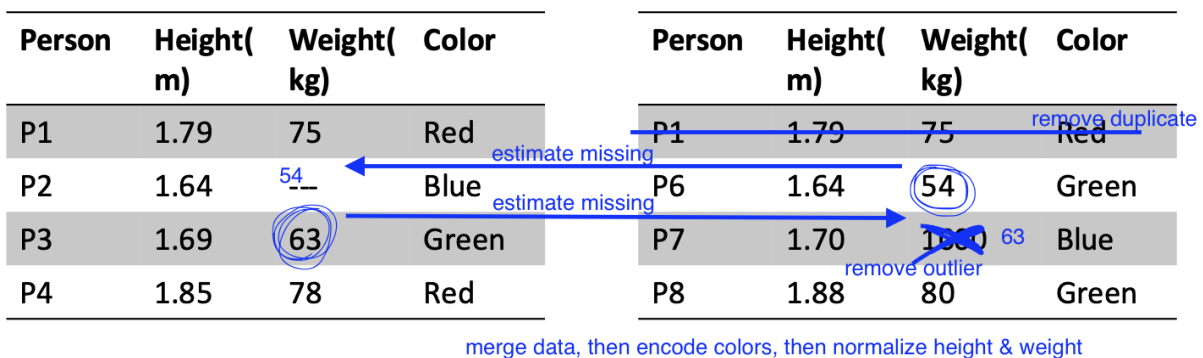
- Data cleaning:
 1. **Denoise** data (if applicable)
 2. Remove **outliers**
 3. Handle **missing** data
 4. Remove **duplicates**
 5. Categorical data **encoding**
 6. Data **normalization**

Data cleaning order affects final results.

If outlier present and data normalization first step → every other data compressed to near 0

Exercise:

Person	Height(m)	Weight(kg)	Color
P1	1.79	75	Red
P2	1.64	54	Blue
P3	1.69	63	Green
P4	1.85	78	Red



Person	Height(m)	Weight(kg)	Color
P1	1.79	75	Red
P6	1.64	54	Green
P7	1.70	100	Blue
P8	1.88	80	Green

merge data, then encode colors, then normalize height & weight

Data exploration

- To identify data quality problems, especially with high quantities of data
- **Summary statistics**
 - Numbers summarizing properties of the data
 - **Frequency, location, spread**
 - e.g. GPA: mean spread, location, standard deviation
 - Can be calculated in a **single pass through**
- Location
 - **Mean** – most common, but sensitive to outliers

$$\bullet \text{ mean}(x) = \frac{1}{m} \sum_{i=1}^m x_i$$

■ Median:

$$\text{median}(x) = \begin{cases} x_{(r+1)} & \text{if } m \text{ is odd, i.e., } m = 2r + 1 \\ \frac{1}{2}(x_{(r)} + x_{(r+1)}) & \text{if } m \text{ is even, i.e., } m = 2r \end{cases}$$

○ Spread

■ Range – diff between max and min

■ Variance or standard deviation – most common, but mean involved so sensitive to outliers

- $\text{variance}(x) = \frac{1}{m-1} \sum_{i=1}^m (x_i - \text{mean}(x))^2$

- $\text{std}(x) = \sqrt{\text{variance}(x)}$

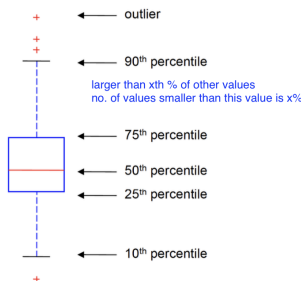
■ Other measures:

• Median absolute deviation (MAD)

- $\text{median}(|x_1 - \text{mean}(x)|, \dots, |x_m - \text{mean}(x)|)$

• Interquartile range

- $x_{75\%} - x_{25\%}$



Percentiles: Given a continuous attribute x and a number p from 0 to 100, the p -th percentile is when $p\%$ of the values of x are **less than** x_p , where x_p is the corresponding value of x at $p\%$

e.g. [1,2,3,4,5,6,7,8,9,10]

→ [4] is 30th percentile, NOT [3]

→ 30% of the values are **less than** x_p → 10 values, so 3 of the values are less than x_p

→ hence x_p is 4

- N values with attribute x , decreasing → $(N \times \frac{1-p}{100})$ -th percentile
- $p = 50$ → x_p is close to median value

Frequency – percentage of time an attribute value appears in the dataset

e.g. for a population, attribute: ‘gender’. Frequency of ‘gender = male’ happens 50% of the time

Mode – most frequently occurring attribute value

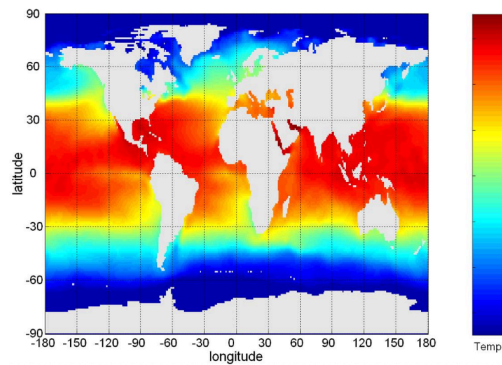
Frequency and mode typically used in **categorical data**

• Visualization

- Converting data to a visual format to allow analysis of data relationships and data characteristics
- Visually present large amounts of information
- Help detect **general patterns and trends**, and **outliers and unusual patterns**

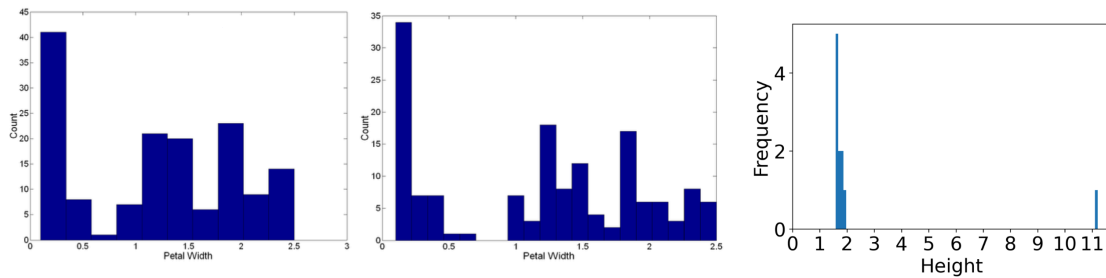
- Techniques:

- Heat map, e.g. sea surface temp.



- Histograms

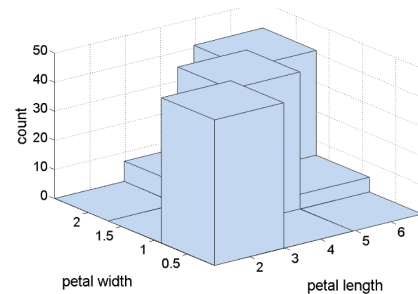
- Show distribution of values
 - Presents data clearly, can quickly identify outliers
 - Height → no. of objects; shape → no. of bins



- 10 bins
- 2 bins every 0.5 interval
- 20 bins
- 4 bins every 0.5 interval

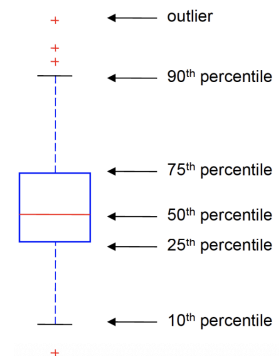
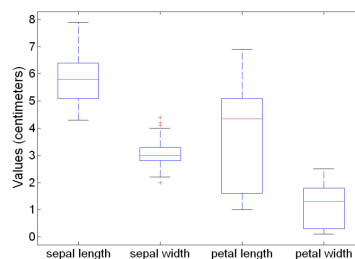
- Two-dimensional histograms

- Joint distribution
 - Shows if there is correlation between two attributes (e.g. comparing width and length)



- Box plots

- Display and compare data distribution



All images and diagrams are taken from BMEG3105 Lecture 5 Slides unless stated otherwise.